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GUILFORD COUNTY DATA ATLAS PROJECT

Measuring Neighborhood Disadvantage & Opportunity: A Literature Review of Indices for Targeting Economic Development in Historically Disinvested Communities

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Abstract

This review synthesizes a now-substantial body of scholarship on neighborhood-level indicators and composite indices used to characterize relative deprivation, economic mobility, socioeconomic conditions, investment and business activity, and community vitality. It situates contemporary American measurement practices within the longer sociological tradition of neighborhood-effects research (Park & Burgess, 1925; Wilson, 1987; Sampson, Raudenbush, & Earls, 1997; Sharkey, 2013) and traces a measurement lineage that runs from the British area-deprivation tradition (Townsend, Phillimore, & Beattie, 1988; Carstairs & Morris, 1991) through the Alkire–Foster counting framework (Alkire & Foster, 2011) that now underwrites the U.S. Census Bureau's Multidimensional Deprivation Index (Glassman, 2019, 2021).

The principal American indices are examined in turn: the Area Deprivation Index and its corrected Reproducible variant (Singh, 2003; Kind & Buckingham, 2018; Petterson, 2023; Reproducible Area Deprivation Index, 2025), the Robert Graham Center Social Deprivation Index (Butler et al., 2013), the Neighborhood Deprivation Index (Messer et al., 2006), the Economic Innovation Group Distressed Communities Index (Economic Innovation Group, 2025), the Opportunity Atlas (Chetty et al., 2018, 2026), the Child Opportunity Index 3.0 (Noelke et al., 2024), the CDC/ATSDR Social Vulnerability Index (Flanagan et al., 2011), the Climate and Economic Justice Screening Tool (White House Council on Environmental Quality, 2022), the Index of Concentration at the Extremes (Krieger et al., 2016), and North Carolina's twin state-level instruments, the statutory County Distress (Tier) Rankings and the new County Economic Vitality Index (NC Department of Commerce, 2025, 2026).

The review attends to units of measure, block group, census tract, ZIP Code Tabulation Area, and county, and to measurement-fit questions including factor-analytic construction, standardization, the modifiable areal unit problem, and the small-area reliability of American Community Survey estimates. An expanded treatment is given to the literature on investment, business activity, and the neighborhood economy, which the deprivation tradition has long underweighted, but which is central to the work of contemporary economic development practitioners. The closing section considers what these indices, taken together, can and cannot tell coalitions working in historically redlined neighborhoods such as those of the North Carolina Piedmont.

Table of Contents

Abstract	ii
1. Introduction	1
2. Theoretical Foundations: Why Neighborhoods Are Measured	2
3. The Historical Substrate: HOLC, Redlining, & Disinvestment	5
4. Indices of Relative Deprivation.....	7
4.1. The Townsend and Carstairs Tradition	7
4.2. The Area Deprivation Index.....	7
4.3. The Robert Graham Center Social Deprivation Index	9
4.4. The Neighborhood Deprivation Index	9
5. Indices of Economic Distress and Community Vitality.....	11
5.1. The Economic Innovation Group Distressed Communities Index	11
5.2. The North Carolina County Distress Rankings (Tier System).....	11
5.3. The North Carolina County Economic Vitality Index.....	12
5.4. Older and Adjacent Instruments.....	13
5.5. UNCG CHCS Neighborhood Market Segmentation Index	13
6. Indices of Opportunity and Intergenerational Mobility	18
6.1. The Opportunity Atlas	18
6.2. The Child Opportunity Index	19
7. Vulnerability Indices: Spanning Economic, Environmental, & Health Risk	20
7.1. The CDC/ATSDR Social Vulnerability Index	20
7.2. The Climate and Economic Justice Screening Tool	21
8. Measures of Segregation & Spatial Polarization	22
9. Investment, Business Activity, and the Neighborhood Economy	23
Newkirk's Neighborhood Economy Index	23
10. Methodological Considerations for the Project Moving Forward.....	24
10.1. Units of Geography: Block, Block Group, Tract, ZCTA, County	24
10.2. Construction Methods: Factor Analysis, Counting, & Equal Weighting.....	25
10.3. Standardization & Choice of Reference Population.....	27
11. Implications for Economic Development Practice in Disinvested Neighborhoods	28
12. Conclusion.....	30
References	32

1. Introduction

The proposition that place matters, that the neighborhood in which a person is raised exerts an independent and durable influence on their life chances, has moved over the past three decades from a contested theoretical claim to something approaching a substantiated fact of American social science (Chetty et al., 2018; Sampson, 2012; Sharkey, 2013, Sills 2021). The implications for economic development practice are far-reaching. If life chances are bound up with geographic context, then community-scale interventions cannot be designed, targeted, or evaluated without a careful empirical account of how neighborhoods differ from one another, how those differences arose, and how they might be changed. Measurement, in other words, is not merely a technical preface to the work of development; it is constitutive of it.

This review surveys the principal indices and composite measures that contemporary scholarship and federal practice have produced to describe neighborhood conditions, with particular attention to those that aim to capture relative deprivation, economic mobility, socioeconomic well-being, investment and business activity, and community vitality. It is organized around five questions:

1. First, what theoretical commitments inform the construction of neighborhood indicators, and how do those commitments shape what the indicators see and do not see?
2. Second, what are the dominant families of indices currently in use, and what are their constituent variables?
3. Third, on what units of geography are they measured, and with what consequences for the kinds of policy targeting they can support?
4. Fourth, what methodological choices, factor analysis, principal components analysis, simple equal-weight averaging, the Alkire–Foster counting approach for multidimensional poverty, undergird the composite scores that policymakers and practitioners ultimately consume?
5. Fifth, and most importantly for practice: what should a coalition working in a historically redlined neighborhood actually do with these tools?

I argue that the contemporary American measurement landscape is best understood not as a single ascendant index but as a layered ecology of complementary instruments, each carrying particular conceptual commitments and particular blind spots. The Distressed Communities Index foregrounds business dynamism and labor-market attachment. The Opportunity Atlas highlights intergenerational outcomes. The Social Vulnerability Index places center variables related to disaster preparedness. The Multidimensional Deprivation Index treats poverty as a failure to achieve real freedoms and opportunities, not just low income. The Area Deprivation Index emphasizes health services research applications. The North Carolina Tier system, statutorily constrained to four county-level variables, foregrounds the political feasibility of a stable annual allocation rule. The Index of Concentration at the Extremes highlights the racialized economic polarization of small areas. Because each instrument is purpose-built, no single index should be

treated as a master rank. The practitioner's task is to triangulate, to read indices against one another and against the lived knowledge of community partners, and to remain attentive to what the chosen unit of geography conceals as well as to what it reveals.

A secondary argument structures the review's treatment of investment and business activity. The dominant family of neighborhood indices is descended from public-health geography and is overwhelmingly concerned with the characteristics of residents (their poverty rates, educational attainment, housing conditions) rather than with the economic activity that places sustain. The Distressed Communities Index is unusual in including two business indicators; beyond these, the standard deprivation literature is largely silent on the questions that matter most for economic development: the density of locally owned businesses, the flow of investment capital into and out of a neighborhood, patterns of anchor institution procurement, and the conditions of the physical commercial corridor. The recent emergence of a neighborhood-economy literature (Porter, 1995; Theodos, Edmonds, & González-Hermoso, 2020; Parilla & Liu, 2018; Initiative for a Competitive Inner City, 2020) and of practitioner-built frameworks such as Newkirk's Neighborhood Economy Index (Newkirklytics, 2026) represents an important methodological frontier, and this review treats that body of work as a parallel measurement enterprise rather than as an adjunct to the deprivation tradition.

2. Theoretical Foundations: Why Neighborhoods Are Measured

The intellectual genealogy of contemporary neighborhood indices runs through four convergent traditions. Any honest accounting of this genealogy must acknowledge its most consequential and most frequently overlooked precursor. W.E.B. Du Bois's *The Philadelphia Negro* (1899/1996), based on a systematic door-to-door enumeration of more than 2,500 Black households in Philadelphia's Seventh Ward conducted between 1896 and 1897, constitutes the first rigorous neighborhood-level social scientific study in the United States. Du Bois combined survey enumeration, hand-drawn spatial maps of household conditions, statistical tabulation, and structural analysis to demonstrate that the concentrated disadvantage of Black Philadelphians was the product of racial discrimination and institutional exclusion rather than individual pathology. This was a methodological and theoretical claim that the neighborhood-effects literature would not fully recover for another nine decades. His insistence that the neighborhood be treated as the unit of analysis, and that conditions within it be explained by structural forces external to its residents, makes *The Philadelphia Negro* the direct, if rarely credited, ancestor of the traditions described below.

The first tradition of neighborhood analytics most often cited is the Chicago School urban ecology of Park and Burgess (1925), which treated the city as a system of differentiated zones whose social characteristics could in principle be mapped, compared, and explained through their position in a broader competitive ecology. The second is the relative deprivation tradition, formalized by Merton and Kitt (1950) in their reanalysis of Stouffer's American soldier studies and elaborated by Runciman (1966) in his account of egoistic and fraternalistic deprivation, which insisted that consequential grievances arise not from absolute conditions but from comparison

with relevant reference groups. The third is the British public-health tradition of small-area deprivation measurement, codified by Townsend, Phillimore, and Beattie (1988) and elaborated through successive generations of the Index of Multiple Deprivation. The fourth is the late-twentieth-century American neighborhood-effects literature, inaugurated by Wilson's (1987) *The Truly Disadvantaged* and given empirical depth by the Project on Human Development in Chicago Neighborhoods (Sampson et al., 1997; Sampson, 2012).

Wilson's (1987) account of the post-industrial inner city argued that the spatial concentration of joblessness, residential segregation, and social isolation produced distinctive neighborhood effects that operated beyond the sum of individual deprivations. Sampson, Raudenbush, and Earls (1997) gave Wilson's claim a tractable empirical form in their multilevel analysis of 343 Chicago neighborhoods, demonstrating that the relationship between concentrated disadvantage and violent crime was substantially mediated by what they called collective efficacy or the combination of social cohesion and informal social control. Crucially for the measurement literature that followed, Sampson and colleagues constructed concentrated disadvantage as a factor-analytic latent variable indexing¹ the joint loadings of poverty, public assistance receipt, unemployment, female-headed households, racial composition, and child density. This factor-analytic operationalization, rather than any single variable, has since become a workhorse construct in urban sociology and the proximate ancestor of many of the indices reviewed below (Sampson, Sharkey, & Raudenbush, 2008).

For the city of Chicago and the U.S. as a whole, representing more than 65,000 census tracts, only one linear factor could be reliably extracted in each decade through principal components analysis (Sampson, 2014).

Massey and Denton (1993), writing in the same period, advanced a complementary measurement agenda focused on residential segregation. Their five-dimensional framework, evenness, exposure, concentration, centralization, and clustering (Massey & Denton, 1988), made clear that segregation is not a single quantity but a family of related geometric properties of the spatial distribution of groups. Each dimension has a corresponding family of statistical measures:² the dissimilarity index for evenness; the isolation and interaction indices for exposure; the Delta or relative concentration index; the proportion of a group's population residing within the central city for centralization; and the spatial proximity or absolute clustering indices for clustering. Their work, alongside the historical scholarship of Rothstein (2017) on the federal government's role in producing residential segregation, provides the political backdrop against which every contemporary neighborhood index must be read: American neighborhoods differ not because they evolved organically but because they were produced through a long sequence of public and private decisions, of which the Home Owners' Loan Corporation maps of the 1930s are the best-known instrument.

¹ See <https://www.qualtrics.com/articles/strategy-research/factor-analysis/> and <https://www.statisticssolutions.com/free-resources/directory-of-statistical-analyses/factor-analysis/>

² See <https://www.census.gov/topics/housing/housing-patterns/guidance/appendix-b.html>

15 Neighborhood-Effect Mechanisms

Broad Categories from George C. Galster, *The Mechanism(s) of Neighborhood Effects*

1. Social-Interactive Mechanisms

-  1 Social Contagion
-  2 Collective Socialization
-  3 Social Networks
-  4 Social cohesion and control
-  5 Competition
-  6 Relative Deprivation
-  7 Parental Mediation

2. Environmental Mechanisms

-  1 Exposure to Violence
-  2 Physical Surroundings
-  3 Toxic Exposure

3. Geographical Mechanisms

-  1 Spatial Mismatch
-  2 Public Services

4. Institutional Mechanisms

-  1 Stigmatization
-  2 Local Institutional Resources
-  3 Local Market Actors

Source: Galster, George C. (2010). *The Mechanism(s) of Neighborhood Effects: Theory, Evidence, and Policy Implications*.

Sharkey's (2013) *Stuck in Place* added a generational dimension to this measurement project, demonstrating that the racial neighborhood inequalities documented by Wilson and Massey and Denton have proven extraordinarily persistent across cohorts. Galster (2012), in a frequently cited synthesis, enumerated fifteen distinct mechanisms by which neighborhood context plausibly shapes individual outcomes, from peer effects and collective socialization to local institutional resources, environmental exposures, and access to networks of opportunity. The proliferation of mechanisms is significant for measurement: any composite index that hopes to capture neighborhood disadvantage must select, weight, and aggregate proxies for processes that operate through quite different causal channels.

A fifth theoretical thread, less often acknowledged in American debates, deserves explicit mention. The Multidimensional Deprivation Index released by the U.S. Census Bureau (Glassman, 2019, 2021) is built on the Alkire–Foster (2011) counting framework, which is itself a formal operationalization of Sen's (1985) capabilities approach. The Sen tradition treats deprivation not as a lack of income but as a curtailment of substantive freedoms: the freedoms to be educated, healthy, adequately housed, securely employed, and integrated into one's community. This conceptual move matters because it disciplines the choice of indicators. Dimensions are selected because they index distinct capabilities, and the index is constructed in a way that explicitly counts overlapping deprivations within persons or households rather than averaging away the joint distribution. Where Sampson-style factor analysis privileges the latent dimension behind correlated indicators, the Alkire–Foster approach privileges the joint incidence of deprivations across distinct dimensions of well-being. Both are defensible as they each answer different questions, and a layered analytic system should be capable of accommodating both.

3. The Historical Substrate: HOLC, Redlining, & Disinvestment

Before turning to contemporary indices, it is essential to acknowledge that the most consequential American “neighborhood index” of the twentieth century was not constructed for analytic or planning purposes but for the explicit allocation of credit. Between 1935 and 1940, the Home Owners' Loan Corporation produced graded residential security maps for roughly 250 cities, classifying neighborhoods on an A-through-D scale in which Black, immigrant, and working-class areas were systematically downgraded (Connolly et al., 2018; Jackson, 1985; Rothstein, 2017). Neighborhoods graded A or B were marked in green and blue and deemed suitable for mortgage lending; neighborhoods graded C and especially D were marked in yellow and red and were treated as high-risk areas in which lending was discouraged. The grading criteria explicitly included racial and ethnic composition. The Mapping Inequality project at the University of Richmond's Digital Scholarship Lab (Nelson et al., 2023) has digitized these maps and made them available alongside georeferenced descriptions of the racialized criteria the appraisers used,

transforming what had been an obscure archival source into one of the most widely used spatial datasets in contemporary urban research.³

The empirical literature on the long-run effects of HOLC grades is now extensive and largely converges on a finding of persistent influence. Aaronson, Hartley, and Mazumder (2021), using a boundary discontinuity design⁴ that compares neighborhoods on opposite sides of HOLC grade lines, found that the maps caused statistically and substantively significant reductions in home ownership, depressed house values and rents, and increased racial segregation for decades after their issuance. Subsequent work has connected redlining to contemporary disparities in school quality, life expectancy, environmental exposures, and intergenerational mobility. Faber (2020) reports that HOLC-appraised cities became and remained more racially segregated than comparable un-appraised cities in subsequent decades. Nardone and colleagues (2020) document persistent associations between HOLC grades and contemporary asthma, heart disease, and life expectancy, mediated by environmental exposures and housing quality. The National Community Reinvestment Coalition (2018, 2022) has documented that, on average, three-quarters of HOLC-graded D neighborhoods remain low-to-moderate income and disproportionately non-white today.

Some methodological caution is warranted. More recent research has emphasized that disinvestment patterns predated the HOLC maps and that the maps' direct influence on private lending was uneven (Fishback, Rose, Snowden, & Storrs, 2022; Michney & Winling, 2020). Michney and Winling (2020), in particular, show that HOLC itself made a non-trivial number of loans to African American borrowers and that the strict redlining narrative obscures meaningful variation in the agency's actual lending practices. The mainstream view, however, treats HOLC grading as the most legible single instrument of a much broader regime of racialized credit rationing that produced today's spatial distribution of disinvestment, a regime that included the Federal Housing Administration's underwriting practices, racially restrictive covenants, urban renewal, highway routing, public housing siting, and successive waves of predatory lending (Connolly, 2014; Massey & Denton, 1993; Rothstein, 2017; Taylor, 2019).

For the purposes of this review, the HOLC literature performs two functions. It establishes the historical specificity of contemporary neighborhood disadvantage, the patterns being measured were produced, not found, and it furnishes an additional spatial overlay that economic development practitioners increasingly incorporate alongside contemporary indices. Practitioners working in the urban Piedmont of North Carolina, for example, can map the late-1930s HOLC grades of Winston-Salem and Greensboro against the contemporary distribution of distress scores and observe the durability of the geography. That overlay is itself a small piece of analysis; it is also a piece of evidence for any equity-focused argument about which neighborhoods deserve targeted investment.

³ Also being conducted in Winston-Salem by Dr. Russ Smith, see <https://www.wfdd.org/2025-01-27/project-mapping-past-racial-covenants-in-forsyth-county-gets-the-community-involved>

⁴ See https://rdpackages.github.io/references/Cattaneo-Titiunik-Yu_2026_ESWC.pdf

4. Indices of Relative Deprivation

4.1. The Townsend and Carstairs Tradition

The contemporary American small-area deprivation literature owes a methodological and conceptual debt to British public-health geography. Townsend and colleagues' (1988) *Health and Deprivation: Inequality and the North* combined four census variables, unemployment, non-car ownership, non-home ownership, and overcrowding, into a single deprivation score that could be calculated for any small geography. The variables were standardized as z-scores after log-transforming the skewed unemployment and overcrowding measures, and the four standardized values were summed unweighted to produce the composite.

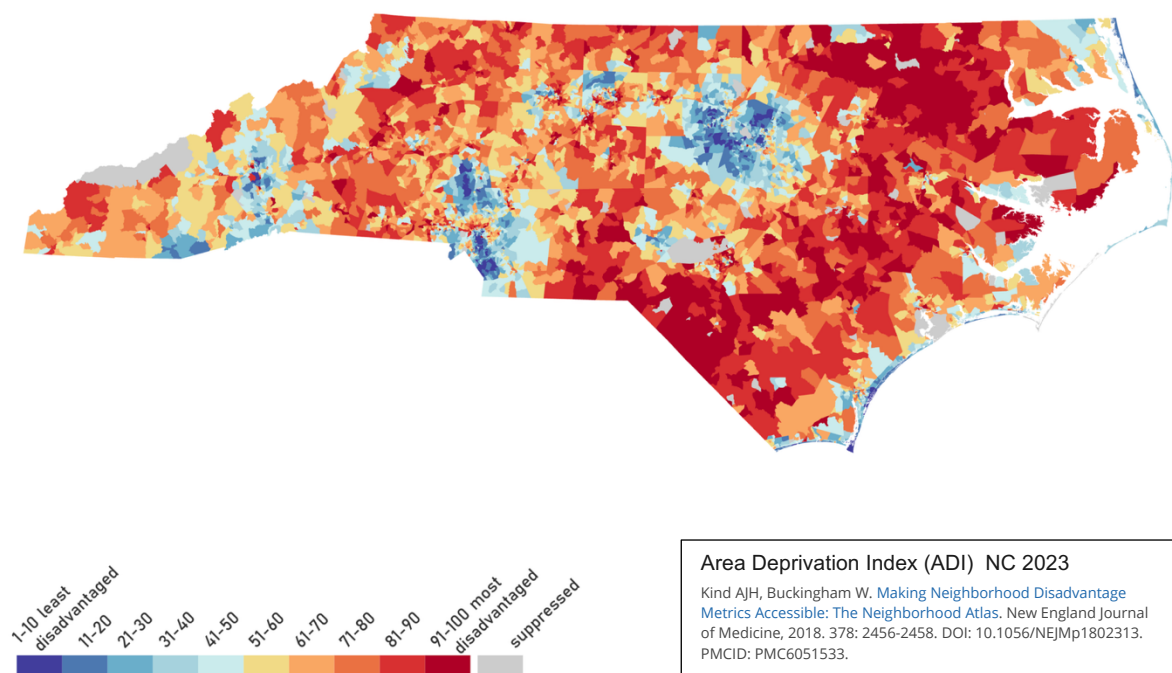
Carstairs and Morris (1989, 1991), working contemporaneously in Scotland, developed a parallel index using unemployment, overcrowding, non-car ownership, and low social class, again summed as standardized z-scores. Successive iterations of the Index of Multiple Deprivation (IMD), now published by the U.K. Ministry of Housing, Communities, and Local Government, expanded the framework to seven weighted domains and made the IMD a routine tool of central and local government resource allocation (Smith et al., 2015). The American adaptations described below all bear the imprint of this tradition: small-area, census-derived, multidimensional, and oriented toward the allocation of public resources.

Two methodological commitments inherited from the British lineage should be noted. The first is the use of equal weighting after standardization: the original Townsend and Carstairs indices treat each component as an equally important manifestation of an underlying construct rather than asserting a priori that some components matter more. The second is the willingness to use indices whose interpretation is explicitly relative: a Townsend or Carstairs score has no absolute meaning but locates an area in the distribution of areas at a given point in time. The Index of Multiple Deprivation departed from this minimalist tradition by introducing weighted domains and by drawing on administrative data in addition to the Census. The American indices reviewed below straddle this divide, with some preserving the parsimonious Townsend logic and others adopting the multi-domain weighted architecture of the IMD.

4.2. The Area Deprivation Index

Among American instruments, the Area Deprivation Index (ADI) has achieved the widest adoption in health services research and is the most explicit heir to the Townsend approach. Originally constructed by Singh (2003) using 1990 Decennial Census data, the ADI applies principal components analysis to seventeen indicators spanning four conceptual domains, income, education, employment, and housing quality, and combines them into a single weighted composite. Kind and colleagues (2014; Kind & Buckingham, 2018) updated the index using American Community Survey data and made it publicly available through the University of Wisconsin's Neighborhood Atlas, where it is published at the census block group and nine-digit ZIP code levels with both national percentile ranks and state-specific deciles.

The ADI has been endorsed by the U.S. Department of Health and Human Services' Office of the Assistant Secretary for Planning and Evaluation (Office of the Assistant Secretary for Planning and Evaluation, 2022) for use in adjusting Medicare payments for social risk, and it appears in a steadily expanding body of biomedical research relating neighborhood disadvantage to outcomes ranging from thirty-day rehospitalization to opioid-related mortality. It has been incorporated into payment models administered by the Centers for Medicare and Medicaid Services, including the Accountable Care Organization Realizing Equity, Access, and Community Health model and the Medicare Shared Savings Program (East Tennessee State University Rural Health Research Center, 2024).



Despite this institutional adoption, the ADI has come under significant methodological criticism. Petterson (2023), in a closely argued analysis published in *Health Affairs Scholar*, demonstrated that the Neighborhood Atlas implementation applies Singh's 1990 tract-level scoring coefficients to contemporary block-group data without standardizing the input variables, with the result that the resulting score is dominated by a single high-variance indicator (median home value). Hannan and colleagues (2023), writing in *Health Affairs proper*, raised related concerns and argued for a more transparent and reproducible implementation.

The Stanford Socioeconomic Position Indices group has subsequently developed a corrected Reproducible Area Deprivation Index (ReADI) that re-estimates factor loadings for each ACS release using contemporary data rather than retaining the 1990 weights, applies proper population weighting, and addresses several other threshold and imputation issues identified in the methodological literature (Reproducible Area Deprivation Index, 2025). These methodological

cautions do not invalidate the ADI but they do counsel against treating its block-group rankings as a precise instrument. For analysts adopting the ADI family, the ReADI rather than the Neighborhood Atlas product is the methodologically preferable choice; in either case, the analytic case for triangulation across measures is reinforced.

4.3. The Robert Graham Center Social Deprivation Index

The Social Deprivation Index (SDI), developed by Butler and colleagues (2013) at the Robert Graham Center for Policy Studies in Family Medicine and Primary Care, was constructed explicitly to inform the geographic allocation of primary care resources. Like the ADI, the SDI is factor-analytic in construction. It combines seven census indicators, the share of residents in poverty, the share with fewer than twelve years of education, the share of single-parent households, the share in renter-occupied housing, the share in overcrowded housing, the share of households without a car, and the share of non-employed adults under age sixty-five, into a composite scored from one to one hundred (Butler et al., 2013; Robert Graham Center, 2019). The Center publishes annual updates at four geographies: county, census tract, ZIP Code Tabulation Area, and Primary Care Service Area.

The SDI has been widely used in studies of access to primary care, multimorbidity, and racial disparities in health outcomes (Bazemore et al., 2015; Mier et al., 2024; Phillips, Liaw, & Bazemore, 2016). Its seven-variable parsimony is both a strength and a limitation: the index is easy to communicate and replicable across years, but its narrow set of indicators omits dimensions, labor force participation among prime-age adults, housing cost burden, environmental exposures, that other instruments include. Researchers comparing the SDI to the ADI generally find moderate-to-high correlations between the two, suggesting that the choice between them often matters less for substantive findings than for the specific policy or clinical question to which the index will be applied. For analysts seeking a peer-reviewed, theoretically grounded, and openly available tract-level deprivation measure, the SDI remains a methodologically defensible choice and is in many respects easier to defend than the ADI.

4.4. The Neighborhood Deprivation Index

Messer and colleagues' (2006) Neighborhood Deprivation Index (NDI) deserves mention as a methodologically rigorous, broadly used, but less institutionalized alternative. Constructed via principal components analysis on a set of socioeconomic indicators including income, education, housing, occupational structure, and unemployment, the NDI was developed in the context of perinatal epidemiology and has since been applied across a wide range of health outcomes. It is typically calculated at the census tract level, scored as a continuous z-statistic, and reported either as a continuous variable or in quartiles. Like the SDI, the NDI exemplifies the factor-analytic family of deprivation indices and is in many respects a methodological cousin of Sampson and colleagues' (1997) concentrated disadvantage construct. The persistent finding across NDI applications, that a single principal component reliably accounts for most of the shared variance among the indicators, is consistent with Sampson et al.'s (2008) demonstration of the same

property for concentrated disadvantage and supports a broader conclusion of this review: area-based deprivation indices are remarkably robust to indicator-set perturbations because their underlying socioeconomic indicators are typically highly inter-correlated.

4.5. The Census Multidimensional Deprivation Index

The Multidimensional Deprivation Index (MDI), released by the U.S. Census Bureau in 2019 and revised several times since, represents the most theoretically distinct of the American deprivation indices (Glassman, 2017, 2019, 2020, 2021, 2024). Where the ADI, SDI, and NDI proceed by factor analysis or principal components analysis on a set of correlated indicators, the MDI is constructed using the Alkire and Foster (2011) dual-cutoff counting approach.⁵ The index identifies individuals as deprived in six dimensions, standard of living, education, health, economic security, housing quality, and neighborhood quality, using indicator-specific cutoffs, and then aggregates a household across dimensions using a second cutoff that specifies how many dimensions of deprivation are required for a household to be classified as multidimensionally deprived (Glassman, 2021). The output is then summarized in three statistics familiar from the multidimensional poverty literature: the headcount ratio H , the intensity of poverty A , and the adjusted headcount ratio $M0$ that combines them (Alkire & Foster, 2011).

The MDI is conceptually attractive for at least three reasons. First, by counting deprivations within households rather than averaging indicator means across geographies, it preserves information about the joint distribution of disadvantage, information that factor-analytic indices systematically discard. Second, because it is built from individual-level American Community Survey microdata, it can be disaggregated by race, ethnicity, age, and household composition in ways that area-level indices cannot. Third, its dimensional structure is explicit and theoretically grounded in Sen's capabilities framework, making the value choices that drive the composite score legible to non-technical audiences. Its principal limitation for neighborhood-targeted economic development work is geographic: the headline MDI estimates are released at national, regional, and state levels rather than at the tract or block group. Glassman (2020) has explored alternative neighborhood-quality measures available at smaller geographies, and Glassman (2022) developed a county-level MDI using five-year ACS estimates, but tract-level MDI publication is not yet feasible without restricted-data access through Federal Statistical Research Data Centers. Practitioners interested in multidimensional small-area measurement can, however, adapt the Alkire–Foster framework using ACS small-area files, as Dhongde, Pattanaik, and Xu (2019) and others have done.

⁵ More at https://ophi.org.uk/sites/default/files/OPHIWP086_Ch5.pdf

5. Indices of Economic Distress and Community Vitality

5.1. The Economic Innovation Group Distressed Communities Index

Where the deprivation indices above are descended from public-health geography, the Distressed Communities Index (DCI), produced annually by the Economic Innovation Group since 2015, is descended from regional economic development and labor-market analysis (Economic Innovation Group, 2017, 2025). The DCI combines seven equally weighted indicators, the share of adults age twenty-five and older without a high school diploma, the housing vacancy rate, the share of prime-age adults not in work, the poverty rate, the ratio of local median income to state median income, the change in employment over the most recent five years, and the change in the number of business establishments over the same period, and translates them into a distress score from zero (most prosperous) to one hundred (most distressed) (Economic Innovation Group, 2022).

The DCI's methodology is more straightforward than that of the factor-analytic indices: each geography is ranked on each of the seven indicators, the ranks are averaged with equal weights, and the resulting average is itself converted to a percentile rank. Geographies are then sorted into five quintile categories, prosperous, comfortable, mid-tier, at risk, and distressed, designed for accessible policy communication (Economic Innovation Group, 2025). The index is calculated at three levels: ZIP code (covering approximately 26,000 areas with at least 500 residents and 99 percent of the U.S. population), county, and congressional district.

Two features of the DCI deserve emphasis for the economic development practitioner. First, in contrast to most deprivation indices, the DCI includes two explicitly dynamic indicators, change in employment and change in business establishments, that are intended to capture trajectory as well as level. A neighborhood that is poor but gaining jobs and business activity scores differently from one that is poor and losing both. This is conceptually important: a place-based development strategy is concerned not only with current conditions but with momentum, and the DCI is among the few national indices to attempt momentum measurement explicitly.

Second, the DCI's use of the ZIP code as its primary geography is both its accessibility advantage and its principal methodological limitation. ZIP codes are postal delivery routes, not census-defined statistical areas; their boundaries shift, they vary enormously in population, and they aggregate neighborhoods that may be qualitatively different. The Economic Innovation Group (2022) explicitly acknowledges that ZIP codes should be considered approximations of geographies and communities rather than exact representations of them. Krieger and colleagues (2002) demonstrated empirically that, for social-epidemiologic outcomes, ZIP-code-based measures consistently obscured inequalities that were clearly visible at the census tract level.

5.2. The North Carolina County Distress Rankings (Tier System)

Several state governments have developed formal economic distress measurement systems used to allocate development incentives and grant program eligibility. North Carolina's system,

established in 1987 and codified at North Carolina General Statute 143B-437.08, ranks the state's one hundred counties annually based on four economic factors: the average unemployment rate, the median household income, the percentage growth in population, and the adjusted property tax base per capita (North Carolina Department of Commerce, 2025). Each county is ranked on each factor, the four ranks are summed to produce a county rank sum (with a possible range of 4 to 400), and counties are then sorted into three tiers by a statutorily mandated 40-40-20 distribution: the forty most distressed counties are designated Tier 1, the next forty are Tier 2, and the twenty least distressed are Tier 3.

The Tier designations determine eligibility and match requirements for a portfolio of state economic development programs, including the One North Carolina Fund, the Job Development Investment Grant program, building reuse grants, and rural infrastructure investments (North Carolina Department of Commerce, 2025). The system is politically durable in part because the fixed 40-40-20 distribution provides a stable annual allocation rule, but it has been subject to recurring critique. NCGrowth and ncIMPACT (2023), in a study commissioned by the North Carolina Collaboratory, reported that stakeholders frequently did not understand the system's mechanics, that its influence on actual business location decisions appeared modest, and that the forced distribution produced anomalies in which counties classified as Tier 3 contained substantial sub-county pockets of distress that were obscured by the county-level aggregate. The critique is especially salient for practitioners working at the neighborhood scale: the Triad counties of Forsyth and Guilford, which contain the historically redlined neighborhoods of Winston-Salem and Greensboro, are routinely classified in the less distressed tiers despite containing census tracts that would qualify as distressed under any reasonable threshold of the EIG Distressed Communities Index. The Tier system, in other words, is an instrument of state-level political economy rather than a neighborhood-level diagnostic, and it should not be used as the latter.

5.3. The North Carolina County Economic Vitality Index

The County Economic Vitality Index, introduced in 2026 by the Labor and Economic Analysis Division of the North Carolina Department of Commerce, is intended to address some of the limitations of the Tier system by providing a transparent and comparable measure of county economic performance that benchmarks against a national reference rather than against other North Carolina counties (North Carolina Department of Commerce, Labor and Economic Analysis Division, 2026).

The Vitality Index uses four indicators (income, educational attainment, labor force participation, and the share of adults with at least a high school diploma) treated equally without weighting. Regression analyses reported by the developers indicate that 2010 Vitality Index scores correlate positively with subsequent rates of job, business establishment, and population growth between 2011 and 2024, with each ten-point increase in the 2010 score associated with approximately ten percentage points of additional employment growth and fourteen percentage points of additional establishment growth over the following fourteen years.

The Vitality Index is conceptually analogous to the Distressed Communities Index but is calibrated against national rather than state benchmarks and is published only at the county level. Its introduction reflects a broader pattern in state economic development analysis: a recognition that the statutory tier system is poorly suited to the analytic and communicative work that the agency increasingly needs to perform, but a recognition also that statutory measures cannot easily be modified. The Vitality Index is therefore best understood as a parallel, non-statutory descriptive tool intended to inform analysis and communication rather than program eligibility. Like the DCI, it remains too coarse for the tract- and block-group-level analysis required for neighborhood-scale economic development planning, but it provides a useful state-and-national reference against which county-level conditions can be compared. For the Guilford County Data Atlas, both NC measures provide important regional context: the Tier system establishes the political-allocation environment in which the project operates, while the Vitality Index permits county-level benchmarking against national peers.

5.4. Older and Adjacent Instruments

Several other instruments deserve brief mention. The Hardship Index, originally developed by Nathan and Adams (1976) and applied to Chicago neighborhoods in the 1970s and refreshed periodically since, combined six standardized indicators, crowded housing, poverty, unemployment, dependency, low educational attainment, and income inequality, into a single zero-to-one-hundred score (Nathan & Adams, 1989). The Hardship Index remains in use as a local-government tool in Chicago and a small number of other jurisdictions. The Urban Institute's neighborhood typologies (Tatian & Snow, 2018) and the Brookings Metropolitan Policy Program's various left-behind places framings (Hendrickson, Muro, & Galston, 2018) operate at a higher level of aggregation and provide useful framing for regional policy but are less suitable for tract-level targeting. The Initiative for a Competitive Inner City (2020) has developed an inner-city business database that tracks establishment density, employment, and ownership characteristics at the tract and ZIP code levels, providing one of the few publicly available data sources oriented specifically toward neighborhood-economy measurement.⁶

5.5. UNCG CHCS Neighborhood Market Segmentation Index

The neighborhood market segmentation index (Sills 2016) constructed by the UNCG Center for Housing and Community Studies for the High Point Core City (later utilized in Greensboro, Rocky Mount, and Baton Rouge Louisiana) study belongs to a well-established family of composite area-based instruments designed to rank small geographies along a dimension of housing market strength and distress. Its immediate intellectual ancestor is the framework developed by Alan Mallach (Center for Community Progress) and applied in studies of Newark, New Jersey, and Youngstown, Ohio, where a small set of housing market indicators was used to classify neighborhoods for the purpose of differentially targeting redevelopment investment.

⁶ Interactive map at

https://smallbizbarriers.smallbusinessmajority.org/?metro=24660&chart_metro=24660&chart_state=37

The UNCG CHCS adaptation extended Mallach's original seven-to-ten-variable housing market model by incorporating sociodemographic, labor force, and physical parcel-condition indicators, producing a seventeen-variable composite that spans housing economics, population characteristics, and on-the-ground property conditions. The conceptual ambition of this extension is sound: housing market stability is not reducible to property values and vacancy alone, and the added indicators capture the social ecology of distress that shapes and reflects market conditions. The result is an instrument that sits at the intersection of the classical housing submarket typology literature (Goodman and Thibodeau 1998; Jones, Leishman, and Watkins 2005) and the neighborhood-level deprivation indices that have become standard in public health and planning research (Butler et al. 2013; Singh 2003).

The seventeen indicators retained in the final index are drawn from four distinct source categories. The first is primary parcel-level assessment data collected directly by CHCS researchers in 2014 using Loveland Technologies remote survey software, covering over 15,000 parcels across 32 block groups in nine census tracts. This category contributes the percent vacant lots and percent substandard structure indicators. The second is secondary administrative and commercial data: ESRI-projected median home values based on the 2010 Decennial Census, Home Mortgage Disclosure Act (HMDA) loan origination counts for 2014, and Guilford County Tax Assessor delinquency records for 2015. The third is American Community Survey five-year estimates for the 2009-2013 period, which furnish the majority of the sociodemographic and housing tenure indicators: owner occupancy, residential mobility, mortgage holding, educational attainment, household income, poverty, food stamp receipt, and age structure. The fourth is local law enforcement data: calls for service involving violent crimes per one thousand residents, drawn from High Point Police Department records for 2014 matched to Census 2010 population denominators.

The initial pool of candidate indicators was substantially larger as the study documentation references more than seventy social, demographic, market, and community variables compiled at the block group level, supplemented by more than twenty parcel-level indicators. This pool was reduced through an iterative process involving bivariate screening and reliability analysis. Specifically, Pearson product-moment correlations were computed between all candidate variables. Variables that did not achieve statistically significant correlations with the emerging composite were removed, as were redundant measures when multiple indicators tapped the same underlying construct and only the strongest retained.

At each reduction step, Cronbach's coefficient alpha was calculated to assess the internal consistency of the remaining item set, and the process was repeated until a reliable, non-redundant set of seventeen indicators was achieved. This iterative reliability-guided reduction is methodologically more disciplined than the expert-judgment variable selection used in many published deprivation indices and reflects a genuine effort to produce a psychometrically coherent composite rather than an ad hoc aggregation of available data.

To account for the fact that the seventeen retained indicators are measured on heterogeneous scales (as percentages, dollar values, counts, and rates) each was standardized to a common metric before aggregation. Standardization was accomplished by computing z-scores against the distribution of values observed across all thirty-two High Point Core City block groups. A z-score transformation expresses each block group's value on a given indicator as the number of standard deviations it lies above or below the mean of the reference population, yielding a dimensionless quantity with mean zero and standard deviation one. This is the same normalization procedure used by the British Townsend and Carstairs deprivation indices (Townsend, Phillimore, and Beattie 1988; Carstairs and Morris 1991), and it ensures that no single indicator dominates the composite merely because of the magnitude of its raw units.

Following standardization, each continuous z-score was recoded to a five-point ordinal scale corresponding to the five market tiers: 1 (Stable), 2 (Functional), 3 (Constrained), 4 (Weak), and 5 (Extremely Weak). The seventeen recoded values were then summed without differential weighting to produce a summative neighborhood index score. By construction, a block group in which every indicator received the most favorable score would receive a total of 17; one in which every indicator received the most distressed score would receive a total of 85. The actual distribution of scores observed in the High Point Core City ranged from 24 (Block Group 137.4, classified Stable) to scores in the low-to-mid seventies for the most distressed block groups. The five-tier classification applied to these summative scores provides the operational basis for the differentiated investment recommendations at the heart of the study.

Several properties of this methodology compare favorably with alternatives in common use. The integration of primary parcel-level assessment data with secondary administrative and survey data is particularly distinctive. The overwhelming majority of published neighborhood deprivation indices including the Area Deprivation Index, Social Deprivation Index, Distressed Communities Index, and Social Vulnerability Index rely exclusively on secondary data sources. The CHCS parcel assessment introduces on-the-ground physical evidence of housing conditions that no remotely assembled dataset can replicate: direct observation of vacant structures, building deterioration, fire damage, and abandonment. This is especially consequential for housing market segmentation specifically, where the physical condition of the housing stock is a causal driver of market values and not merely a symptom of socioeconomic conditions. The iterative reliability-testing procedure provides an empirical basis for indicator retention that is more transparent and reproducible than the expert-judgment or committee consensus methods used to select variables in many governmental indices. The five-tier classification scheme, calibrated to the local context rather than imposing a national threshold, is well-suited for the decision support purpose of the study: it provided planners with an actionable, intuitively communicable typology without the false precision of a continuous score.

Four limitations of the current methodology merit acknowledgment, both for transparency with readers of this report and to inform any future iterations of the index.

First, the reference population against which z-scores are computed consists of only thirty-two block groups. Z-score standardization assumes that the mean and standard deviation of the reference distribution are stable estimates of the underlying population parameters. With $n = 32$, these estimates are sensitive to the presence of outlier block groups: a single extreme value can substantially shift the mean and compress or expand the standard deviation, distorting every block group's standardized score (a flaw that may be reduced with a county or statewide matrix). The practical consequence is that the composite index rankings are particularly sensitive to the composition of the study area itself. Had the study boundary been drawn to include or exclude a few additional block groups, the rankings of many middle-range neighborhoods could change materially. This is a specific instantiation of the modifiable areal unit problem (Openshaw 1984) and is an inherent limitation of any locally normed composite. It is, notably, the same limitation explicitly flagged in the study's own documentation: "the index is internally cohesive and valid, but is not useful in comparisons to other locales." This underscores that the composite is best understood as a local ordinal ranking instrument rather than as a measure of absolute deprivation.

Second, the conversion of continuous z-scores to five-point ordinal categories before summation introduces a categorical artifact into what had been continuous interval data. Once a z-score is collapsed to a 1-5 integer, the information about its precise location within that tier is discarded. A block group at the very top of the "Constrained" range (just below "Weak") contributes an identical 3 to the composite as one at the very bottom of that range (just above "Functional"), despite representing materially different underlying conditions. The summative index then treats these five-point codes as if they were interval, which is the standard assumption behind arithmetic addition but is not strictly warranted for ordinal categories. An alternative construction would retain the continuous z-scores and summing them directly prior to any tier classification thus preserving more information, reduce the categorical artifact, and produce a composite with better statistical properties. The tier classification could then be applied to the continuous composite score for communication purposes without requiring it to be embedded in the computational procedure.

Third, and relatedly, the equal-weighting assumption that governs the summation treats every retained indicator as making an equivalent contribution to the underlying construct. This is theoretically coherent when there is no *a priori* basis for differential weighting, as is the case here, and it preserves transparency. However, the Pearson correlation matrix reported in the study documentation reveals that the seventeen indicators vary considerably in their degree of inter-correlation with the composite. Median home value ($r = .914$ with the loan origination measure, $r = .819$ with median household income) loads much more heavily on the apparent latent dimension of neighborhood distress than, for example, percent foreign-born (which has a more complex correlation structure). Factor-analytic weighting, as used in the Social Deprivation Index (Butler et al. 2013) and the original Area Deprivation Index (Singh 2003), would assign higher weights to indicators that share more variance with the dominant latent dimension, potentially improving the composite's predictive validity. For a study area of thirty-two units, factor analysis is statistically underpowered, which makes the equal-weighting approach a reasonable pragmatic choice; this constraint should be explicitly acknowledged rather than left implicit.

Fourth, the inclusion of the percent non-White population as one of the seventeen composite indicators warrants careful consideration. Race and ethnicity are correlated with neighborhood distress in High Point, as they are in virtually every American city, because of a long history of racially discriminatory housing policy, credit rationing, and urban disinvestment that the study's background section documents in detail. Using racial composition as a component of the distress composite conflates the structural cause of disinvestment with its measurement. It risks implying that a neighborhood's distress score would improve if its racial composition changed, rather than if the underlying conditions of credit access, property investment, and public infrastructure improved. The CDC Social Vulnerability Index has received precisely this critique (Flanagan, Hallisey, Adams, and Lavery 2018), and it applies equally here. A methodologically preferable approach would treat racial and ethnic composition as a contextual variable that structures the interpretation of the composite score rather than as a constituent element of it.

The CHCS High Point index occupies a distinctive position in the landscape of area-based composite instruments reviewed elsewhere in this study. It shares its z-score standardization and equal-weighting logic with the British Townsend index and with Sampson, Raudenbush, and Earls' (1997) concentrated disadvantage construct, both of which demonstrated the methodological utility of standardized summation for small-area measurement. It differs from these precedents in incorporating primary observational data, in applying an intermediate categorical transformation prior to summation, and in operating on a reference population of thirty-two units rather than the thousands of tracts or block groups typical in national applications.

Relative to the Economic Innovation Group's Distressed Communities Index (Economic Innovation Group 2025), the CHCS index shares a local housing and labor market orientation but differs in using z-scores rather than percentile ranks, in incorporating physical condition data absent from any national index, and in being calibrated to a local rather than national reference population. The DCI's inclusion of trajectory indicators — change in employment and change in establishments over five years — is a feature the CHCS index lacks; the CHCS instrument describes current conditions at a single point in time and cannot distinguish a neighborhood that is declining from one that has stabilized at a low level.

Relative to the Area Deprivation Index (Singh 2003; Kind and Buckingham 2018), the CHCS index is more theoretically coherent in its standardization procedure. The Neighborhood Atlas implementation of the ADI has been shown to apply 1990-era factor score coefficients to unstandardized contemporary data, producing a composite dominated by median home value (Petterson 2023; Hannan et al. 2023). The CHCS approach, by contrast, computes z-scores contemporaneously against the study-area distribution for each indicator, avoiding this specific error. The corrected Stanford Reproducible ADI (Reproducible Area Deprivation Index 2025), which re-estimates factor loadings against current data with proper standardization, represents the methodologically preferable national instrument for the block-group geography at which both it and the CHCS index operate.

Relative to the Guilford County Neighborhood Vitality Index (GNVI) now under development through the Guilford County Data Atlas, the CHCS High Point index represents an important methodological precedent. Both share a z-score normalization convention, a block-group unit of analysis, a multidomain indicator structure, and a summative aggregation logic. The GNVI departs from the CHCS model in several significant respects: it uses the full Guilford County tract distribution as its reference population (substantially larger and more stable than thirty-two block groups), it assigns differential domain weights grounded in Newkirk's production-capacity framework rather than treating all indicators equally, and it explicitly separates population-based deprivation measurement from neighborhood-economy benchmarking rather than merging them into a single composite. The CHCS High Point study's five-tier market typology and its integration of parcel-level primary data remain directly relevant precedents for the Atlas design.

6. Indices of Opportunity and Intergenerational Mobility

6.1. The Opportunity Atlas

Perhaps the most consequential development in American neighborhood measurement of the past decade has been the Opportunity Atlas,⁷ constructed by Chetty, Friedman, Hendren, Jones, and Porter (2018) and now published as Chetty et al. (2026) in the *American Economic Review*. The Opportunity Atlas departs from the cross-sectional logic that animates every other index reviewed here. Rather than describing the contemporary characteristics of a neighborhood's adult residents, it follows the cohort of approximately 20.5 million Americans born between 1978 and 1983 from the census tract in which they grew up to their adult outcomes in their mid-thirties. The result is a tract-level estimate of children's expected outcomes, mean adult earnings, incarceration rates, teenage birth rates, and several other measures, conditional on the income, race, and gender of their parents (Chetty et al., 2018, 2026).

The methodological foundation of the Atlas is a set of papers in which Chetty and Hendren (2018a, 2018b) developed a movers-design quasi-experimental identification strategy that compares children whose families moved between counties at different ages, exploiting variation in childhood exposure. The county-level results indicate that each additional year of childhood exposure to a one-standard-deviation higher-mobility county increases adult income by approximately 0.04 standard deviations of the income distribution. Chetty, Hendren, and Katz (2016) extend these findings to the Moving to Opportunity experimental sample, providing one of the few directly causal demonstrations of place effects on long-run outcomes.

The Atlas's headline empirical finding is that intergenerational mobility varies sharply across nearby tracts in ways that are not well predicted by traditional measures of neighborhood opportunity. For children of parents at the twenty-fifth percentile of the household income distribution, the standard deviation of mean adult household income at age thirty-five is approximately \$5,000 across tracts within the same county; tract-level poverty rates, the variable

⁷ Interactive map at: <https://www.opportunityatlas.org/>

most often used in neighborhood-effects research, explain only about half of the variation in adult outcomes (Chetty et al., 2018). Combined with quasi-experimental and experimental evidence from the Moving to Opportunity demonstration and from Chetty and Hendren's (2018a, 2018b) movers design, the Atlas evidence suggests that roughly sixty percent of the cross-neighborhood variation in adult outcomes is causal, that is, attributable to the neighborhood itself rather than to the characteristics of families who sort into it.

For economic development practice, the Atlas is unique in that it measures outcomes that any plausible development intervention would care about, will children raised here flourish economically as adults, rather than the present-day conditions that are themselves means to those outcomes. The Atlas is also distinctive in its construction: the underlying microdata are housed at the Census Bureau and accessed under restricted-use agreements, so the published tract-level estimates are the product of careful disclosure-avoidance work rather than a directly downloadable individual-level file. This restriction has methodological implications for users, who cannot in general replicate the Atlas or interrogate alternative specifications, but it does not diminish the analytic power of the published estimates. Practitioners working in disinvested neighborhoods can use Atlas data to identify, with considerable spatial precision, the tracts in which children growing up in poor families have historically achieved the worst adult outcomes, and to target interventions, whether place-based investments or housing-mobility programs, accordingly.

6.2. The Child Opportunity Index

The Child Opportunity Index (COI), developed at the Heller School at Brandeis University and published through diversitydatakids.org, is the most ambitious effort to date to construct a multidimensional composite measure of the present-day neighborhood conditions relevant to child well-being (Acevedo-Garcia et al., 2014, 2020; Noelke et al., 2024). The current version, COI 3.0, is built from 44 component indicators organized into three domains (education, health and environment, social and economic) and 14 subdomains. The 44 indicators include measures of early childhood education, school quality, post-secondary enrollment, green space and walkability, food environment, environmental exposures, employment, income, wealth, housing quality, residential stability, broadband access, density of local nonprofits, and social capital (Noelke et al., 2024).

COI 3.0 introduced several methodological refinements over its predecessors. The index is constructed at the census block level for both the 2010 and 2020 Decennial Census geographies, with publication at the census tract level for analytic use.⁸ Component indicator processing makes use of contemporary machine-learning tools, including the OpenAI text embeddings used to classify food retailers and the lasso, ridge, and random-forest models used to refine school quality measures (Noelke et al., 2024). The index is published as both an overall Child Opportunity Score

⁸ Access the Guilford County map view at <https://www.diversitydatakids.org/maps/#/explorer/tracts/37081010900/23/13,18,23//x/c/n/5.1.1/36.079/-79.79/10.26/>

(1 to 100) and as domain and subdomain scores, enabling users to interrogate, for example, whether a neighborhood's relatively low overall score reflects poor educational opportunity, poor environmental conditions, or socioeconomic disadvantage. Indicators are weighted not equally but by the strength of their bivariate association with an outcome index derived from the Opportunity Atlas and from CDC PLACES health data, an outcome-anchored weighting strategy that distinguishes the COI from most of the other indices reviewed here.

The COI's principal contribution is its breadth: where the deprivation indices canvassed earlier focus on what residents lack, the COI attempts to measure what neighborhoods provide. For economic development practitioners, the conceptual asymmetry matters. A neighborhood may score poorly on the ADI and well on certain COI subdomains because it has, say, strong educational institutions despite low household income. Another neighborhood may score similarly on the ADI but lack institutional resources entirely. The two profiles call for different interventions. Validation analyses reported by the developers indicate that COI 3.0 outperforms competing composite indices, including its own predecessor COI 2.0, on measures of predictive and equity validity (diversitydatakids.org, 2025). Like the Opportunity Atlas, the COI also provides ZIP code estimates as a courtesy for users whose data systems require them, but the Heller School authors are explicit that the tract-level estimates are the methodologically preferred unit of analysis.

7. Vulnerability Indices: Spanning Economic, Environmental, & Health Risk

7.1. The CDC/ATSDR Social Vulnerability Index

The Social Vulnerability Index (SVI),⁹ developed by Flanagan and colleagues (2011) at the Centers for Disease Control and Prevention's Geospatial Research, Analysis, and Services Program and maintained by the Agency for Toxic Substances and Disease Registry, was initially constructed to support emergency management and disaster preparedness. The current iteration uses sixteen census variables drawn from the American Community Survey five-year estimates, organized into four thematic domains: socioeconomic status (poverty, unemployment, low income, no high school diploma); household composition and disability (age, single-parent households, disability); minority status and language; and housing type and transportation (multi-unit structures, mobile homes, crowding, no vehicle, group quarters) (Centers for Disease Control and Prevention, 2024; Flanagan et al., 2011).

Methodologically, the SVI computes percentile ranks for each of the sixteen indicators at the census-tract level, sums the percentile ranks within each of the four themes, and then computes an overall percentile rank from the sum of the theme totals. The result is published as a tract-level percentile from 0 to 1, with higher values indicating greater vulnerability (Centers for Disease Control and Prevention, 2024). Although the SVI was developed for disaster management, its components are nearly identical to those of conventional social-deprivation indices, and it has

⁹ Interactive map at <https://www.atsdr.cdc.gov/place-health/php/svi/svi-interactive-map.html>

been widely adopted in public-health, economic-development, and equity-screening applications well beyond its original purpose (Cutter, Boruff, & Shirley, 2003).

The recent literature has begun to critique the SVI's reliance on simple summation rather than factor analysis. Jurecka and colleagues (2025), for example, demonstrate that a factor-analytic implementation produces meaningfully different vulnerability rankings and argue for explicit modeling of the latent structure rather than assumed equal weights. Flanagan and colleagues' (2021) reply to an earlier critique by Rufat and colleagues makes a related point in defense of the original approach. A separate critique concerns the inclusion of racial and ethnic composition as a vulnerability variable: this practice has been the subject of recurring methodological debate, since it risks treating racial composition as itself a deficit rather than as a marker of accumulated structural disadvantage (Flanagan, Hallisey, Adams, & Lavery, 2018). The SVI nonetheless retains its institutional position as a default screening tool because of its accessibility, its annual updating, and the documentation provided by CDC/ATSDR. For the purposes of an economic development atlas, the SVI is best treated as a complementary rather than substitutive measure: useful for understanding population characteristics that condition the response to investment, but not itself a measure of economic distress or opportunity.

7.2. The Climate and Economic Justice Screening Tool

The Climate and Economic Justice Screening Tool (CEJST), produced by the White House Council on Environmental Quality under Executive Order 14008, was designed to identify the disadvantaged communities eligible for the prioritized investment specified by the Justice40 Initiative (White House Council on Environmental Quality, 2022). Methodologically, the CEJST takes an unusual hybrid form: it is neither a single composite ranking nor a simple set of thresholds but a binary classification produced by the joint satisfaction of two conditions. A census tract is designated disadvantaged if it exceeds the ninetieth percentile threshold on at least one of approximately three dozen burden indicators, spanning climate change, energy, health, housing, legacy pollution, transportation, water and wastewater, and workforce development, and exceeds the threshold for an associated socioeconomic indicator (typically a measure of low income or low educational attainment) (White House Council on Environmental Quality, 2022). Version 1.0 of the CEJST identified approximately 27,000 disadvantaged or partially disadvantaged tracts.

The CEJST is a useful instrument for several reasons. It explicitly integrates environmental and economic dimensions of disadvantage, recognizing that these dimensions reinforce one another in spatially concentrated ways. It is tract-level rather than ZIP-level. And it was constructed with explicit reference to historical disinvestment, including a HOLC-redlining indicator in its housing burden category. Its principal limitation is the binary classification: a tract is either disadvantaged or it is not, and a tract meeting multiple burden criteria receives the same designation as one meeting only a single criterion (World Resources Institute, 2022).

The Center for Neighborhood Technology (2024) and other researchers have documented cases of tracts narrowly excluded from CEJST eligibility despite obvious cumulative burdens, illustrating

the familiar problem of threshold-based indices producing arbitrary cliffs in eligibility. Practitioners using the CEJST should understand the tool both as a federal funding-allocation instrument and as a methodologically defensible identification of communities for whom multiple lines of evidence suggest sustained disinvestment. Note that public access to the CEJST through the White House was discontinued in January 2025; archived versions and unofficial mirrors remain available, and the underlying methodology can be reconstructed independently using the documented thresholds and the public ACS and EPA source data.

8. Measures of Segregation & Spatial Polarization

Although segregation measures are sometimes treated as a separate analytic enterprise from deprivation indices, the historical and theoretical literature reviewed above suggests that the two should be considered jointly. The Massey and Denton (1988) five-dimensional framework, evenness, exposure, concentration, centralization, and clustering, remains the standard reference for the geometric analysis of group distributions across small areas. Each dimension has a corresponding family of statistical measures: the dissimilarity index for evenness; the isolation and interaction indices for exposure; the Delta or relative concentration index; the proportion of a group's population residing within the central city for centralization; and the spatial proximity or absolute clustering indices for clustering. The 2020 Census data and the corresponding American Community Survey five-year estimates have produced a renewed cycle of analyses showing that segregation along the evenness and exposure dimensions has declined modestly in many metropolitan areas while concentration and clustering have remained more stubborn (Logan & Stults, 2022).

For practitioners interested specifically in the joint distribution of race and economic position, Krieger and colleagues' (2016) Index of Concentration at the Extremes (ICE) provides a particularly accessible instrument.¹⁰ Adapting an approach originally developed by Massey (2001), Krieger and colleagues constructed three variants. The ICE for income takes the difference between the number of high-income and low-income households divided by the total population with known income. The ICE for race takes the difference between the number of non-Hispanic White and the number of Black or Hispanic residents divided by the total population. The ICE for race and income combined takes the difference between high-income non-Hispanic White households and low-income Black or Hispanic households divided by total households. All three variants are scaled from negative one to positive one, with negative values indicating concentration of deprivation, positive values indicating concentration of privilege, and zero indicating mixing (Krieger et al., 2016).

The ICE has been widely adopted in epidemiological and urban-health research and has accumulated an evidence base linking the racialized economic variant in particular to a range of outcomes from preterm birth to firearm violence to police-involved deaths to premature mortality (Feldman, Waterman, Coull, & Krieger, 2015; Krieger et al., 2017; Lange-Maia et al., 2022). Its

¹⁰ See example ArcGIS application <https://www.arcgis.com/apps/dashboards/2aa946018d2f4554a3484d9521a1057d>

analytic advantage over conventional segregation indices is that it produces a single tract-level scalar that can be used directly in regression models without recourse to two-group dissimilarity statistics that are inherently metropolitan rather than neighborhood quantities.

For an economic development practitioner working in a historically redlined neighborhood, the ICE for race and income combined provides a parsimonious summary of the joint racialized economic position of the area in a form that is easily mapped, compared across periods, and connected to historical HOLC grades. The ICE should be regarded as a complement to the deprivation and opportunity indices reviewed above rather than as a substitute for them: it captures the polarization of a neighborhood relative to its metropolitan area, which neither the absolute deprivation indices nor the outcome-based Opportunity Atlas explicitly measures.

9. Investment, Business Activity, and the Neighborhood Economy

The indices reviewed above are overwhelmingly concerned with the population characteristics of places (their poverty rates, their educational attainment, their housing conditions) rather than with the business activity that places sustain. The Distressed Communities Index is unusual in including two business indicators, change in employment and change in establishments, and the County Economic Vitality Index attends to labor force participation. Beyond these, however, the standard family of deprivation indices is largely silent on the questions that matter most for economic development: the density of locally owned businesses, the flow of investment capital into and out of a neighborhood, the patterns of anchor institution procurement, and the conditions of the physical commercial corridor.

This gap is being addressed in a growing literature on the neighborhood economy as an analytic object distinct from neighborhood demographic composition. Porter's (1995) early articulation of the competitive advantage of the inner city emphasized the strategic underutilization of urban markets, urban workforces, and proximity to downtown. Cooke and Rolfe (2006) and Theodos, Edmonds, and González-Hermoso (2020) extended this argument with empirical work on community development financial institutions, demonstrating that capital availability is the binding constraint on small-business growth in many distressed neighborhoods. Parilla and Liu (2018) at the Brookings Institution have developed a framework for inclusive economic development that places anchor institution procurement and locally owned business density at the center of place-based strategies. The Initiative for a Competitive Inner City (2020) has produced an inner-city business database that tracks establishment density, employment, and ownership characteristics at the tract and ZIP code levels. Howard, Klein, and Hart (2021) have separately documented the persistent disparities in small-business credit access that condition this geography.

Newkirk's Neighborhood Economy Index

Newkirk's Neighborhood Economy Index, currently in development as the analytic backbone of the Guilford County Data Atlas, sits squarely within this emerging tradition (Newkirklytics, 2026). The index organizes neighborhood economic conditions around six dimensions: local business density, the entrepreneur pipeline, capital access, anchor institution engagement, economic

infrastructure, and community economic activity, and operationalizes each through benchmark targets per ten thousand residents drawn from the highest-performing comparable neighborhoods. The conceptual move from cross-sectional deprivation measurement to benchmark-based gap analysis is methodologically significant. It shifts the analytic question from how distressed a neighborhood measures to what is missing for this neighborhood to function as a healthy local economy, and it points directly toward intervention design rather than toward ranking and targeting alone.

Several supplementary data sources permit small-area measurement of business activity that the population-based indices cannot. The Home Mortgage Disclosure Act (HMDA) provides annual loan-level data on residential mortgage applications and originations at the census tract level, permitting the construction of neighborhood credit-access measures. Community Reinvestment Act disclosures provide parallel data on small-business lending by regulated depository institutions, although the resolution of CRA data is generally coarser than HMDA. County Business Patterns provides annual establishment, employment, and payroll data at the ZIP-code and county levels, though disclosure-avoidance suppression reduces its usefulness at finer geographies. The Longitudinal Business Database, accessible through Federal Statistical Research Data Centers, provides establishment-level data that can be analyzed at the tract level under restricted-use protocols. The Esri Business Analyst and Data Axle commercial data products provide proprietary establishment-level data that, while imperfect, can be used for tract-level analysis without disclosure restrictions. Sills and Buaka (2023) have used HMDA data to document disparate credit access across North Carolina counties, and the Coalition of Anchors work in Guilford County contemplates a parallel exercise for small-business and corporate-investor lending.

The integration of population-based deprivation indices with neighborhood-economy indices is, accordingly, the methodological frontier most relevant to the Guilford County project. Deprivation indices identify the places in which intervention is most warranted by current population conditions. Neighborhood-economy indices identify the specific economic-system gaps that intervention must close. Used together, they provide both the where and the what of place-based economic development planning. This is the analytic posture that the Guilford County Data Atlas is being designed to support.

10. Methodological Considerations for the Project Moving Forward

10.1. Units of Geography: Block, Block Group, Tract, ZCTA, County

The selection of a unit of geography is the single most consequential methodological decision in neighborhood measurement, and the indices reviewed above illustrate the full range of available choices. The U.S. Census Bureau hierarchy proceeds from block (the smallest unit, typically a city block bounded by streets) through block group (typically 600 to 3,000 residents), to census tract (typically 1,200 to 8,000 residents and designed to approximate a neighborhood), to county subdivisions and counties. The ZIP Code Tabulation Area is a Census Bureau approximation of

U.S. Postal Service ZIP codes constructed from census blocks; ZCTAs vary enormously in population and do not nest cleanly within counties (U.S. Census Bureau, 2020).

The unit-of-geography choice interacts with two well-established methodological problems. The first is the modifiable areal unit problem (MAUP), first identified by Gehlke and Biehl (1934) and given canonical formulation by Openshaw (1984). The MAUP arises because individual-level phenomena that are aggregated to areal units produce different statistical patterns depending on both the scale of aggregation (the scale effect) and the specific way boundaries are drawn at any given scale (the zoning effect). Fotheringham and Wong (1991) demonstrated, in a much-cited empirical analysis, that the MAUP is not merely an abstract concern but can produce substantively different conclusions from the same underlying data. Vogel (2016), in a study using Add Health data, demonstrated that the same substantive relationship between neighborhood disadvantage and adolescent violence appeared at the census tract level but disappeared at the block group level, not because one measurement was correct and the other was not, but because the two measurements were answering subtly different questions. Hipp (2007), in a widely cited paper, made a parallel argument about the level-of-aggregation choice in studies of crime and disorder. The practical implication is straightforward: any single index at any single scale should be interpreted with reference to other indices at other scales, not as a definitive ranking.

The second methodological problem is the small-area reliability of American Community Survey estimates. Because the ACS is a sample rather than a census, its estimates for small geographies, especially block groups and small tracts, carry substantial standard errors, and the five-year estimates required for block group publication smooth across periods in ways that can obscure recent change (Spielman, Folch, & Nagle, 2014; Spielman & Folch, 2015). Block group estimates of, for example, the share of households without vehicle access can carry coefficients of variation in excess of 30 percent in low-population geographies, meaning that two block groups with apparently different point estimates may not in fact be reliably distinguishable. The University of Wisconsin Neighborhood Atlas implementation of the ADI publishes block-group rankings without published standard errors or reliability indicators, a practice that has been one of the principal objects of methodological critique (Petterson, 2023). Practitioners using small-area indices should consult, where possible, the underlying ACS margins of error and treat fine-grained rankings as approximations rather than precise measurements.

10.2. Construction Methods: Factor Analysis, Counting, & Equal Weighting

The composite indices reviewed above are constructed by three principal families of methods. The first, factor analysis or principal components analysis, is used by the ADI, SDI, NDI, and by Sampson and colleagues' (1997) concentrated disadvantage construct. These methods assume that the observed indicators are correlated reflections of one or more latent dimensions, identify the latent dimensions through eigendecomposition¹¹ of the correlation matrix, and produce composite scores using estimated factor loadings as weights. **The principal strength of factor-**

¹¹ See <https://www.datacamp.com/tutorial/eigendecomposition>

analytic construction is that it is empirically disciplined: the weights are derived from the data rather than imposed by the analyst. The principal weakness is that the resulting composite is dominated by whatever underlying dimension explains the most variance, which may or may not align with the dimension the analyst intended to measure (Pettersen, 2023). Factor-analytic indices also assume that the latent structure is constant across geographies and time periods, an assumption that is empirically debatable.

Best-practice recommendations for factor-analytic index construction are well-established in the methodological literature. Costello and Osborne (2005), Fabrigar, Wegener, MacCallum, and Strahan (1999), and Hayton, Allen, and Scarpello (2004) collectively advise that the appropriateness of factor reduction be assessed before its application using the Kaiser-Meyer-Olkin measure and Bartlett's test of sphericity; that the number of retained factors be determined through parallel analysis rather than the Kaiser eigenvalue criterion alone; that factor loadings be reported transparently with attention to communality and cross-loadings; and that the analysis be re-estimated periodically against contemporary data rather than retained indefinitely from the date of original construction (Tabachnick & Fidell, 2019). Few of the institutional implementations of factor-analytic indices in current use satisfy all of these criteria, which is part of what makes the methodological critique of the Neighborhood Atlas ADI so consequential.

The second family is the Alkire–Foster counting framework, used by the Census MDI. As described above, this approach identifies deprivations at the individual or household level using indicator-specific cutoffs and then counts deprivations using a second cutoff to identify the multidimensionally deprived. Aggregate measures are constructed as the headcount ratio (H), the average intensity of deprivation among the deprived (A), and the adjusted headcount ratio ($M0 = H$ multiplied by A). The counting framework preserves information about the joint distribution of deprivations across persons rather than collapsing across them, and it makes value choices about dimensions and weights explicit rather than burying them in factor loadings (Alkire & Foster, 2011).

The third family is equal-weighted percentile averaging, used by the EIG Distressed Communities Index, the County Economic Vitality Index, and, in slightly modified form, by the SVI and the North Carolina Tier system. The approach ranks each geography on each indicator, averages the ranks (or percentile-equivalent transforms), and converts the result back to a percentile or category. **Equal weighting is sometimes criticized as theoretically naive, but it has the considerable virtue of being transparent: every input variable contributes equally, and the resulting index is interpretable to a non-specialist audience.** The disadvantage is that the choice of which indicators to include, and how to define them, drives the resulting score in ways that are not always made explicit in the documentation. The broader composite-indicator literature (Booyesen, 2002; Decancq & Lugo, 2013; Greco, Ishizaka, Tasiou, & Torrisi, 2019; Nardo et al., 2008; OECD & Joint Research Centre, 2008) treats the choice of weighting scheme as itself a substantive methodological commitment that should be made transparently and tested through sensitivity analysis.

Across all three families, the question of validity is harder than it appears. Most validation exercises proceed by demonstrating that a composite index is correlated with some health or socioeconomic outcome of interest. But because virtually every plausible composite is correlated with virtually every plausible outcome at the area level, such validation establishes little more than that the index is measuring something within the broad domain of socioeconomic disadvantage. **Genuine measurement-fit validation requires comparing alternative composites on a common set of outcomes, examining the marginal contribution of each indicator, and, ideally, engaging in confirmatory factor analysis or structural equation modeling against an explicit theoretical model.** Few of the indices reviewed here have been subjected to such validation, although Flanagan and colleagues (2021) and Jurecka and colleagues (2025) provide useful examples for the SVI specifically. Cronbach and Meehl's (1955) classic articulation of construct validity remains the conceptual standard against which validation work should be assessed.

10.3. Standardization & Choice of Reference Population

A final, often neglected methodological question concerns standardization. Most factor-analytic indices require that input variables be standardized (typically as z-scores) before the factor analysis is performed, because the resulting factor loadings are otherwise dominated by whichever variable happens to have the largest absolute variance. The Petterson (2023) critique of the Neighborhood Atlas ADI implementation centers precisely on this issue: the Wisconsin team applied Singh's (2003) tract-level factor score coefficients to unstandardized 2020 block-group data, producing scores in which a single high-variance indicator, median home value, dominates the composite. The substantive consequence is that block groups in expensive metropolitan housing markets receive disadvantage scores that bear little relation to other measures of distress. Hannan and colleagues' (2023) parallel demonstration showed that **more than seventy percent of block groups change deciles when the same procedure is applied with proper standardization, and several hundred block groups move from the least-deprived to the most-deprived decile or vice versa.** This is among the most consequential methodological errors documented in the contemporary deprivation-index literature, and it should counsel caution against treating any factor-analytic composite as a precise instrument absent careful examination of its underlying standardization.

A related question is the choice of reference population for percentile ranking. The Neighborhood Atlas ADI publishes both national percentiles and state-specific deciles, on the reasonable theory that policy applications often require state-relative comparisons. The DCI and SVI publish only national percentiles. The COI publishes both national and metropolitan-area-relative scores. **The choice matters substantively: a Guilford County tract that ranks at the seventieth national percentile of deprivation may rank at the fortieth percentile of North Carolina deprivation, and the policy implications of each ranking are different.** Practitioners should be aware of the reference population for any index they cite and should choose accordingly. For the Guilford

County Data Atlas, the analytic decision could be to report national, state, and county-relative rankings for each index, with explicit signage indicating which is which.

11. Implications for Economic Development Practice in Disinvested Neighborhoods

What, finally, should the practitioner do with this measurement landscape? The literature reviewed above supports seven conclusions of practical importance.

First, **no single index should be treated as a master ranking** of neighborhood need. Each instrument reviewed here was constructed for a particular purpose, with a particular set of conceptual commitments, and reports a particular slice of the underlying social reality. The Distressed Communities Index speaks most clearly to local labor markets and business dynamism. The Area Deprivation Index speaks most clearly to health services research applications. The Multidimensional Deprivation Index speaks most clearly to overlapping capabilities deprivations. The Opportunity Atlas speaks most clearly to long-run intergenerational outcomes. The Child Opportunity Index speaks most clearly to the contemporary opportunity structure facing children. The Social Vulnerability Index speaks most clearly to disaster preparedness and emergency response. The Index of Concentration at the Extremes speaks most clearly to racialized economic polarization. The CEJST speaks most clearly to overlapping environmental and economic burdens. The practitioner who wishes to make a defensible case about neighborhood need should triangulate across these instruments and be prepared to explain what each one is contributing.

Second, **the appropriate unit of geography depends on the question**. For neighborhood-scale economic development (housing repair, weatherization, small-business support, gentle density infill, community development financial institution lending) the census tract is generally the most useful default, balancing the small-area specificity needed for targeting against the sampling reliability needed for credible estimates. Block-group analysis can usefully complement tract-level analysis where the tract is large or internally heterogeneous, but block-group estimates should be read with explicit attention to their ACS margins of error. ZIP code analyses, including the DCI and SDI ZCTA estimates, should be treated as approximations whose internal heterogeneity may obscure the patterns of greatest local interest.

Third, **the county-level instruments**, including the North Carolina Tier system and the new County Economic Vitality Index, **should not be relied upon for neighborhood-level diagnostic work**. They were designed for state-level resource allocation politics and for state-and-national benchmarking, not for the identification of disinvested neighborhoods within metropolitan counties. A practitioner working in the historically redlined East Greensboro neighborhood of Old Asheboro will find no diagnostic purchase in the fact that Guilford County is classified as Tier 3, just as a practitioner working in the Washington Street or Southwest neighborhoods of High Point will find little useful information in the Tier classification of Guilford County. For these

neighborhoods, the tract-level DCI, ADI or ReADI, COI, ICE, and Opportunity Atlas estimates, read together with the historical HOLC overlay, will be considerably more informative.

Fourth, **deprivation indices should be paired with neighborhood-economy measures rather than used in isolation.** Population-based indices identify the places where intervention is warranted by resident conditions; neighborhood-economy frameworks (the Neighborhood Economy Index, the inner-city business database, HMDA and CRA lending data, anchor procurement records) identify the specific economic-system gaps that intervention must close. A diagnostic system that reports only population conditions will tend to produce intervention recommendations that focus on residents' deficits rather than on the structural conditions of the local economy in which they live; a system that reports only neighborhood-economy conditions will tend to ignore the human-capital and social-fabric considerations that ultimately determine whether economic interventions take root. The Guilford County Data Atlas, by integrating both, should be designed to support the more balanced analytic posture.

Fifth, the **historical dimension matters.** The contemporary distribution of distress is the cumulative product of a long sequence of public and private decisions, and the analytic value of the Mapping Inequality HOLC overlay (Nelson et al., 2023) lies precisely in its capacity to render that history legible alongside contemporary indices. Neighborhoods that were graded D or C in 1937 and that still register in the bottom quintile of the Distressed Communities Index in 2025 have not failed to thrive; they have been, in the most literal sense, prevented from thriving. The implication is not merely rhetorical. Economic development interventions in such neighborhoods need not begin from the premise that the area's residents have failed to make appropriate market choices; they can instead begin from the premise that capital has been systematically withheld and that public and philanthropic investment is a form of overdue repair. The framing matters for the design of measurement systems. An index used to guide investment in formerly redlined neighborhoods should not portray those neighborhoods as deficient; it should portray them as the historical sites of disinvestment whose vitality is now to be rebuilt.

Sixth, **quantitative indices should be combined with the local knowledge of community partners.** The published literature on participatory program evaluation and on community-based research has been clear for some time that the residents of disinvested neighborhoods possess detailed and accurate knowledge of their own conditions that area-level census data cannot capture (Israel et al., 2010; Minkler & Wallerstein, 2008). This is not a methodological concession; it is a substantive observation about the limits of secondary data. The HOLC field surveys themselves are a useful historical analogue: those surveys were conducted by appraisers who walked the streets and talked to local realtors, and the resulting grades, however discriminatory in their intent, encoded a great deal of detailed local information that no remote analyst could have constructed from census tables. The contemporary equivalent, neighborhood walks, resident focus groups, partnership with neighborhood associations, remains an indispensable complement to indices. The author's prior work in the Washington Street Historic Business District

(Sills, 2025) and in the High Point Reparations Commission process (Sills, 2023) is built on precisely this premise.

Seventh, and finally, **the appropriate stance toward measurement is one of disciplined humility**. The indices reviewed here are useful approximations. Each can be made to mislead; none should be invoked as definitive. The practitioner who treats them as orienting devices rather than as oracles, who reads them against one another rather than in isolation, and who combines them with attentive listening to the residents of the neighborhoods they describe will be considerably better served than the practitioner who selects a single index because it produces a single number that fits on a single page of a grant application. The neighborhoods being measured deserve more than that, and so does the work.

12. Conclusion

The measurement of neighborhood disadvantage and opportunity has matured considerably over the past two decades. Where the late twentieth century debate centered on whether neighborhood effects existed in any meaningful sense, the contemporary debate is conducted within a substantial methodological literature on how to measure them, at what scale, with what variables, and toward what ends. The indices canvassed here, the Multidimensional Deprivation Index, the Area Deprivation Index and its Reproducible variant, the Social Deprivation Index, the Neighborhood Deprivation Index, the Distressed Communities Index, the North Carolina Tier rankings, the North Carolina County Economic Vitality Index, the UNCG CHCS Market Segmentation approach, the Hardship Index, the Opportunity Atlas, the Child Opportunity Index, the Social Vulnerability Index, the Climate and Economic Justice Screening Tool, and the Index of Concentration at the Extremes, are now joined by a substantial historical literature on the Home Owners' Loan Corporation maps and their long-run effects, by a continuing methodological literature on factor analysis, the Alkire–Foster counting framework, the modifiable areal unit problem, and small-area survey reliability, and by an emerging neighborhood-economy literature centered on business density, capital access, anchor procurement, and corridor activity.

For economic development practitioners working in historically redlined and urban renewal neighborhoods, this literature counsels neither the use of any single index nor the abandonment of measurement in favor of pure qualitative judgment. It counsels instead a layered, triangulated, and theoretically informed approach in which multiple instruments are read against one another, in which the unit of geography is selected to match the question, in which population-based deprivation measures are paired with neighborhood-economy measures, in which historical context is taken seriously, and in which the lived knowledge of community partners is treated as a substantive data source rather than as window-dressing for predetermined conclusions. The neighborhoods that need investment most urgently are not difficult to identify. They appear at the deprived extremes of every credible index. They appear on the HOLC maps of the 1930s. And they appear in the lived experience of their residents. The work that remains is not measurement but the appropriate alignment of resources, policy, and institutional commitment with what the

measurements already reveal. The Guilford County Data Atlas should be designed to support that alignment.

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